Amazon Networking Services

Overview

This document introduces **Amazon cloud networking**, which enables communication between cloud resources and with external networks. Networking is foundational to designing secure, scalable AWS architectures.

Topics covered include:

1. **Computer Network Basics** – Core networking concepts such as IP addresses, subnets, and CIDR notation.
2. **Amazon Virtual Private Cloud (VPC)** – The virtual network where AWS resources (for example, EC2) communicate.
3. **AWS Network Security** – Firewalls and access controls used to protect cloud resources.
4. **AWS Direct Connect** – Dedicated private connectivity between on-premises data centers and AWS.
5. **Amazon DNS (Route 53)** – Scalable domain name resolution and traffic routing.
6. **Amazon CDN (CloudFront)** – Low-latency global content delivery.

Understanding these services builds a strong foundation for designing and operating secure AWS cloud environments.

Reviewing Computer Network Basics

A **computer network** consists of interconnected devices that share resources and exchange data. A **subnet** is a logical subdivision of a network, created to improve performance, security, and manageability.

Each subnet is identified by a **network address**, and every device within it is assigned a unique **IP address** to enable communication.

IP Address

An **IP (Internet Protocol) address** uniquely identifies a device on a network and enables data exchange. Two versions exist:

- **IPv4** – 32-bit addresses written as four decimal numbers (for example, `172.217.9.36`)
- **IPv6** – 128-bit hexadecimal addresses

This chapter focuses on IPv4.

Figure 3.1 illustrates a sample website and its IPv4 address.

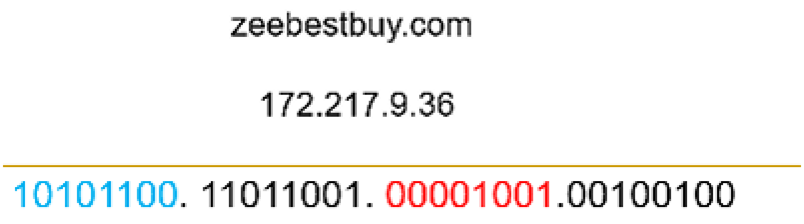


Figure 3.1 – An IP address

An IPv4 address contains **32 bits (4 bytes)**, allowing a theoretical total of 2^{32} addresses. Since a single network cannot practically contain all these addresses, networks are divided into subnets using **CIDR**.

CIDR

Classless Inter-Domain Routing (CIDR) defines IP address ranges and subnet sizes. CIDR notation consists of an IP address followed by a slash (`/`) and the number of fixed network bits.

Example: `172.217.9.0/24`

- First 24 bits: network portion
- Remaining 8 bits: host portion

This subnet contains **256 IP addresses** (`172.217.9.0` to `172.217.9.255`).

Figure 3.2 shows common CIDR ranges and address capacities.

Subnet Mask Hierarchy

Subnet Mask	CIDR	Binary Notation	Available Addresses Per Subnet
255.255.255.255	/32	11111111.11111111.11111111.11111111	1
255.255.255.254	/31	11111111.11111111.11111111.11111110	2
255.255.255.252	/30	11111111.11111111.11111111.11111100	4
255.255.255.248	/29	11111111.11111111.11111111.11111000	8
255.255.255.240	/28	11111111.11111111.11111111.11110000	16
255.255.255.224	/27	11111111.11111111.11111111.11100000	32
255.255.255.192	/26	11111111.11111111.11111111.11000000	64
255.255.255.128	/25	11111111.11111111.11111111.10000000	128
255.255.255.0	/24	11111111.11111111.11111111.00000000	256
255.255.254.0	/23	11111111.11111111.11111110.00000000	512
255.255.252.0	/22	11111111.11111111.11111100.00000000	1024
255.255.248.0	/21	11111111.11111111.11111000.00000000	2048
255.255.240.0	/20	11111111.11111111.11110000.00000000	4096
255.255.224.0	/19	11111111.11111111.11100000.00000000	8192
255.255.192.0	/18	11111111.11111111.11000000.00000000	16384
255.255.128.0	/17	11111111.11111111.10000000.00000000	32768
255.255.0.0	/16	11111111.11111111.00000000.00000000	65536
255.254.0.0	/15	11111111.11111110.00000000.00000000	131072
255.252.0.0	/14	11111111.11111100.00000000.00000000	262144
255.248.0.0	/13	11111111.11111000.00000000.00000000	524288
255.240.0.0	/12	11111111.11110000.00000000.00000000	1048576
255.224.0.0	/11	11111111.11100000.00000000.00000000	2097152
255.192.0.0	/10	11111111.11000000.00000000.00000000	4194304
255.128.0.0	/9	11111111.10000000.00000000.00000000	8388608
255.0.0.0	/8	11111111.00000000.00000000.00000000	16777216

Figure 3.2 – CIDR notation

CIDR enables flexible network design and efficient IP address allocation.

The Internet

The **internet** is a global system of interconnected networks that exchange data using standardized protocols such as TCP/IP.

Key components include:

- **Routers** – Forward traffic between networks using route tables
- **Switches** – Connect devices within a network using MAC addresses

Each subnet is associated with exactly **one route table**, which determines how traffic is forwarded.

Modern networks, like compute systems, have evolved from physical hardware to **software-defined virtual networks**, such as Amazon VPC.

Understanding Amazon Virtual Private Cloud

An **Amazon VPC** is a logically isolated virtual network in the AWS cloud. It allows full control over IP addressing, subnets, routing, and security.

Key benefits include:

1. **High Availability** – Regional scope with subnets spanning Availability Zones
2. **Flexibility** – Custom CIDR ranges, routing, and network topology
3. **Isolation** – Separation of workloads (for example, production vs development)
4. **Cost Efficiency** – No charge for creating or using VPCs
5. **Security** – Subnet-level and instance-level firewalls
6. **Hybrid Connectivity** – Integration with on-premises networks via VPN or Direct Connect

---

Part One – Creating a VPC with Subnets

This section demonstrates VPC and subnet creation using both the AWS Management Console and AWS CloudShell.

Creating a VPC and a Subnet Using the AWS Console

Create a VPC named `my-vpc1` with CIDR `10.10.0.0/16` and no IPv6 block.

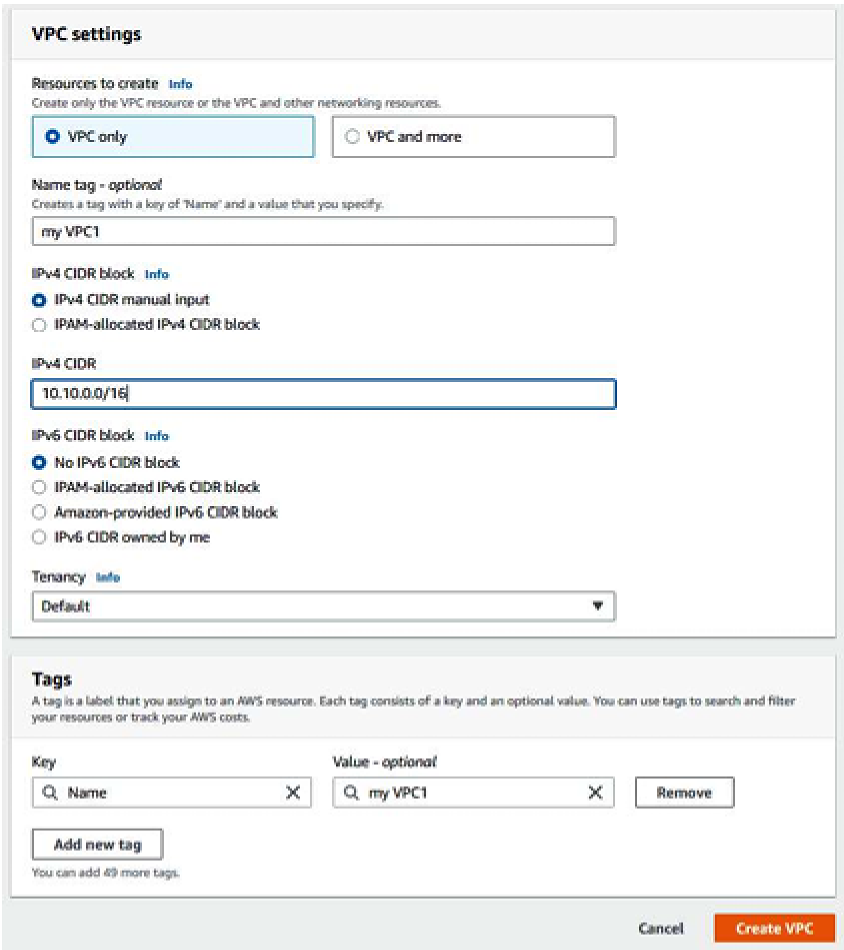


Figure 3.3 – VPC settings

Create a subnet named `public-subnet` with CIDR `10.10.1.0/24`.

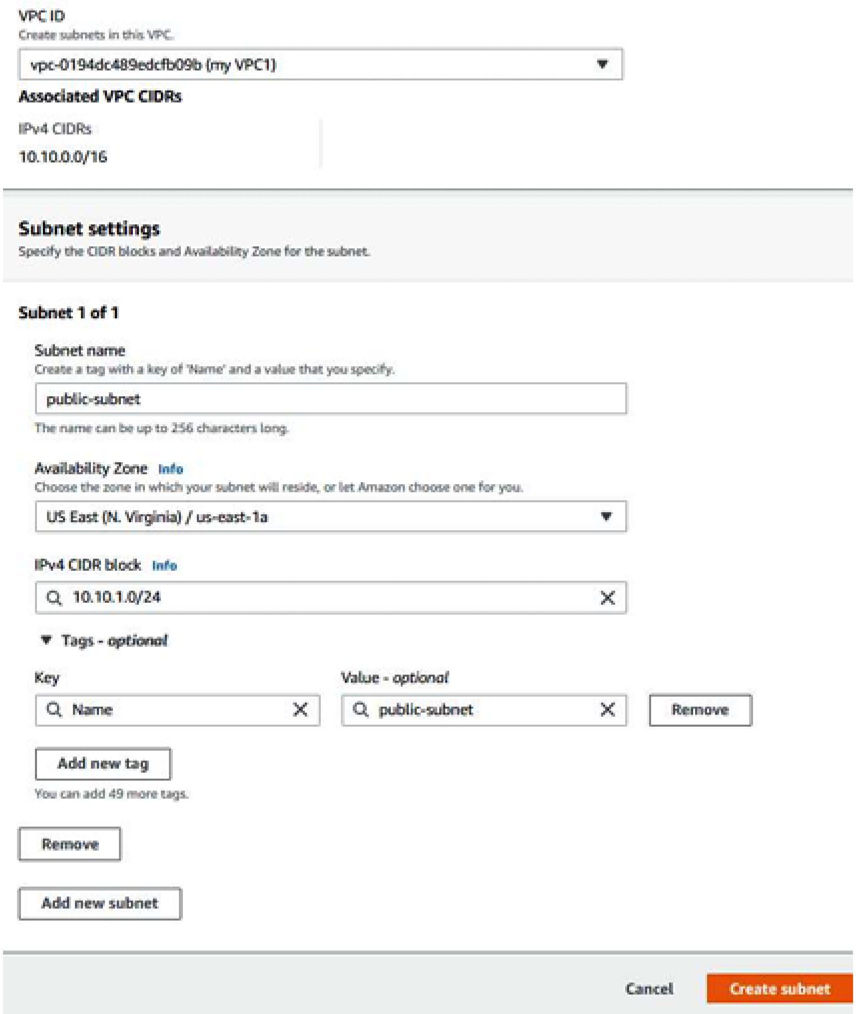


Figure 3.4 – Subnet settings

Creating an AWS Cloud Network Using CloudShell

AWS CloudShell provides CLI-based resource management.

1. Create VPC1

```
aws ec2 create-vpc --cidr-block 10.10.0.0/16 \
--tag-specifications ResourceType=vpc,Tags='[{Key=Name,Value="VPC1"}]'
```

```
[cloudshell-user@ip-10-2-126-119 ~]$ aws ec2 create-vpc \
>     --cidr-block 10.10.0.0/16 \
>     --tag-specifications ResourceType=vpc,Tags='[{Key=Name,Value="VPC1"}]'
{
    "Vpc": {
        "CidrBlock": "10.10.0.0/16",
        "DhcpOptionsId": "dopt-f8579e9d",
        "State": "pending",
        "VpcId": "vpc-0e6043d6091faa7cd",
        "OwnerId": "317332158300",
        "InstanceTenancy": "default",
        "Ipv6CidrBlockAssociationSet": [],
        "CidrBlockAssociationSet": [
            {
                "AssociationId": "vpc-cidr-assoc-04116e8e5e98b256f",
                "CidrBlock": "10.10.0.0/16",
                "CidrBlockState": {
                    "State": "associated"
                }
            }
        ],
        "IsDefault": false,
        "Tags": [
            {
                "Key": "Name",
                "Value": "VPC1"
            }
        ]
    }
}
```

Figure 3.5 – Create VPC1

2. Create Public and Private Subnets

```
aws ec2 create-subnet --vpc-id vpc-xxxx \
--cidr-block 10.10.1.0/24 \
--tag-specifications ResourceType=subnet,Tags='[{Key=Name,Value="public-subnet"}]'

aws ec2 create-subnet --vpc-id vpc-xxxx \
--cidr-block 10.10.2.0/24 \
--tag-specifications ResourceType=subnet,Tags='[{Key=Name,Value="private-
subnet"}]'
```

```
[cloudshell-user@ip-10-2-126-119 ~]$ aws ec2 create-subnet \
>     --vpc-id vpc-0e6043d6091faa7cd \
>     --cidr-block 10.10.2.0/24 \
>     --tag-specifications ResourceType=subnet,Tags='[{Key=Name,Value="private-subnet"}]'
{
    "Subnet": {
        "AvailabilityZone": "us-west-2a",
        "AvailabilityZoneId": "usw2-az1",
        "AvailableIpAddressCount": 251,
        "CidrBlock": "10.10.2.0/24",
        "DefaultForAz": false,
        "MapPublicIpOnLaunch": false,
        "State": "available",
        "SubnetId": "subnet-0328a7940791d9fc0",
        "VpcId": "vpc-0e6043d6091faa7cd",
        "OwnerId": "317332158300",
        "AssignIpv6AddressOnCreation": false,
        "Ipv6CidrBlockAssociationSet": [],
        "Tags": [
            {
                "Key": "Name",
                "Value": "private-subnet"
            }
        ],
        "SubnetArn": "arn:aws:ec2:us-west-2:317332158300:subnet/subnet-0328a7940791d9fc0",
        "EnableDns64": false,
        "Ipv6Native": false,
        "PrivateDnsNameOptionsOnLaunch": {
            "HostnameType": "ip-name",
            "EnableResourceNameDnsARecord": false,
            "EnableResourceNameDnsAAAARecord": false
        }
    }
}
```

Figure 3.6 – Create subnets

3. Launch EC2 Instances

One instance is launched in each subnet.

```
[cloudshell-user@ip-10-6-43-95 ~]$ aws ec2 run-instances --image-id ami-0ef0b498cd3fe129c --count 1 \
> --instance-type t2.micro --subnet-id subnet-0328a7940791d9fc0 \
> --key-name mywestkp --region us-west-2  --no-associate-public-ip-address
{
    "Groups": [],
    "Instances": [
        {
            "AmiLaunchIndex": 0,
            "ImageId": "ami-0ef0b498cd3fe129c",
            "InstanceId": "i-0288c5d94570afe62",
            "InstanceType": "t2.micro",
            "KeyName": "mywestkp",
            "LaunchTime": "2023-02-14T01:56:33+00:00",
            "Monitoring": {
                "State": "disabled"
            },
            "Placement": {
                "AvailabilityZone": "us-west-2a",
                "GroupName": "",
                "Tenancy": "default"
            },
            "PrivateDnsName": "ip-10-10-2-11.us-west-2.compute.internal",
            "PrivateIpAddress": "10.10.2.11",
```

Figure 3.7 – Create instances

Figure 3.8 shows the resulting VPC architecture.

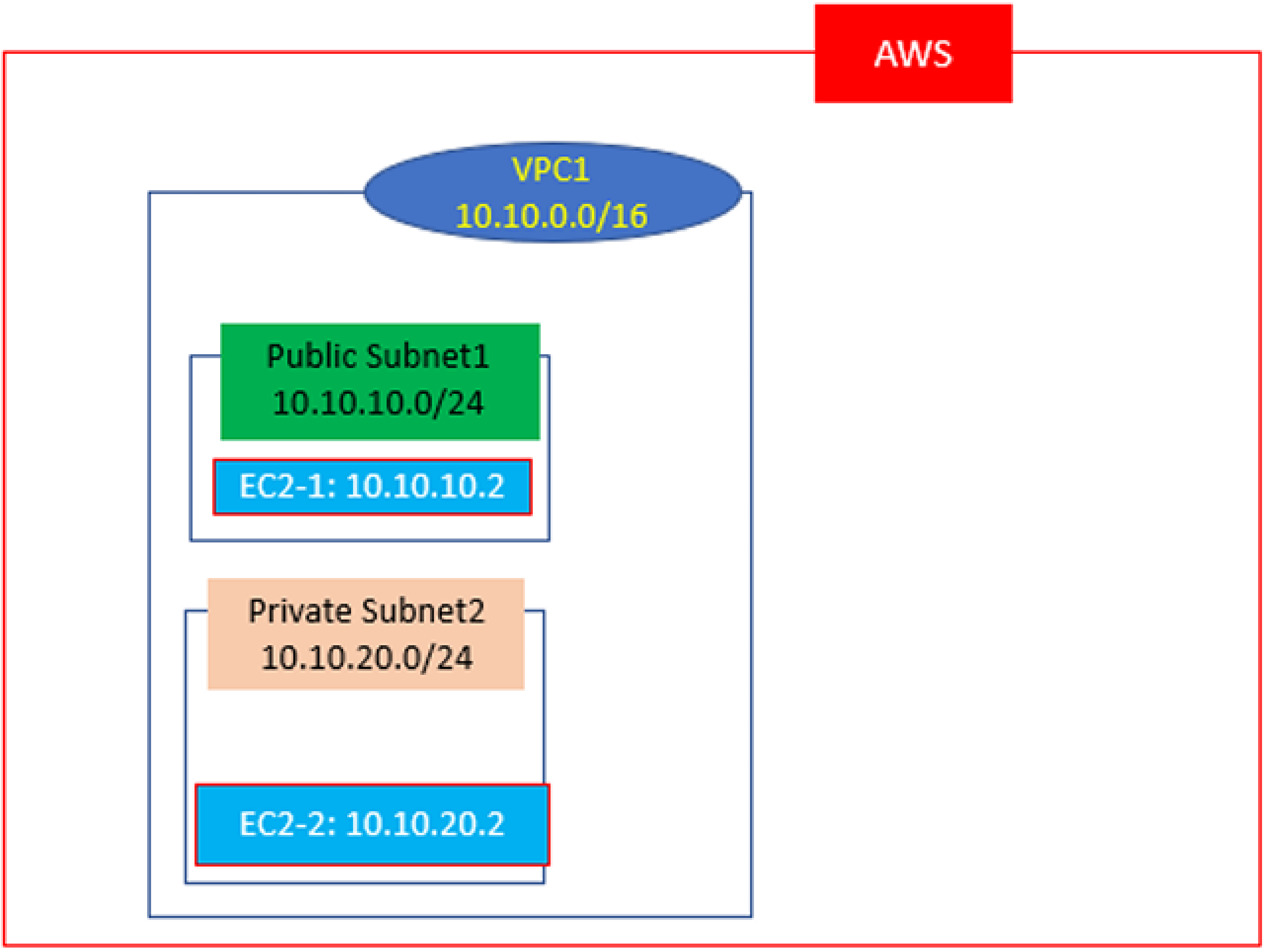


Figure 3.8 – VPC network

Accessing EC2 Instances from the Internet

Public subnet instances require an **Internet Gateway (IGW)** and appropriate routing.

1. Create and Attach an Internet Gateway

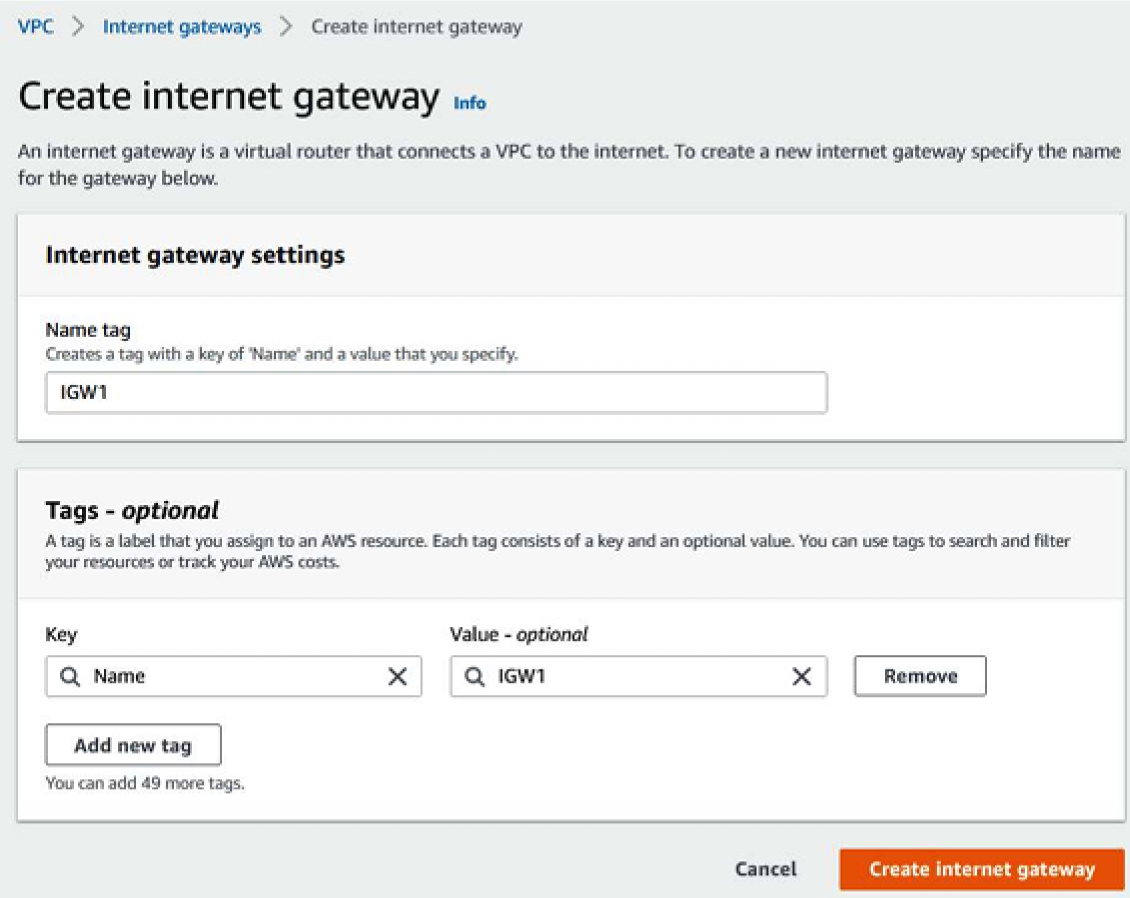


Figure 3.9 – Create IGW1

Figure 3.10 – Attach IGW1 to VPC1

2. Update Route Table

Add a default route (`0.0.0.0/0`) pointing to the IGW.

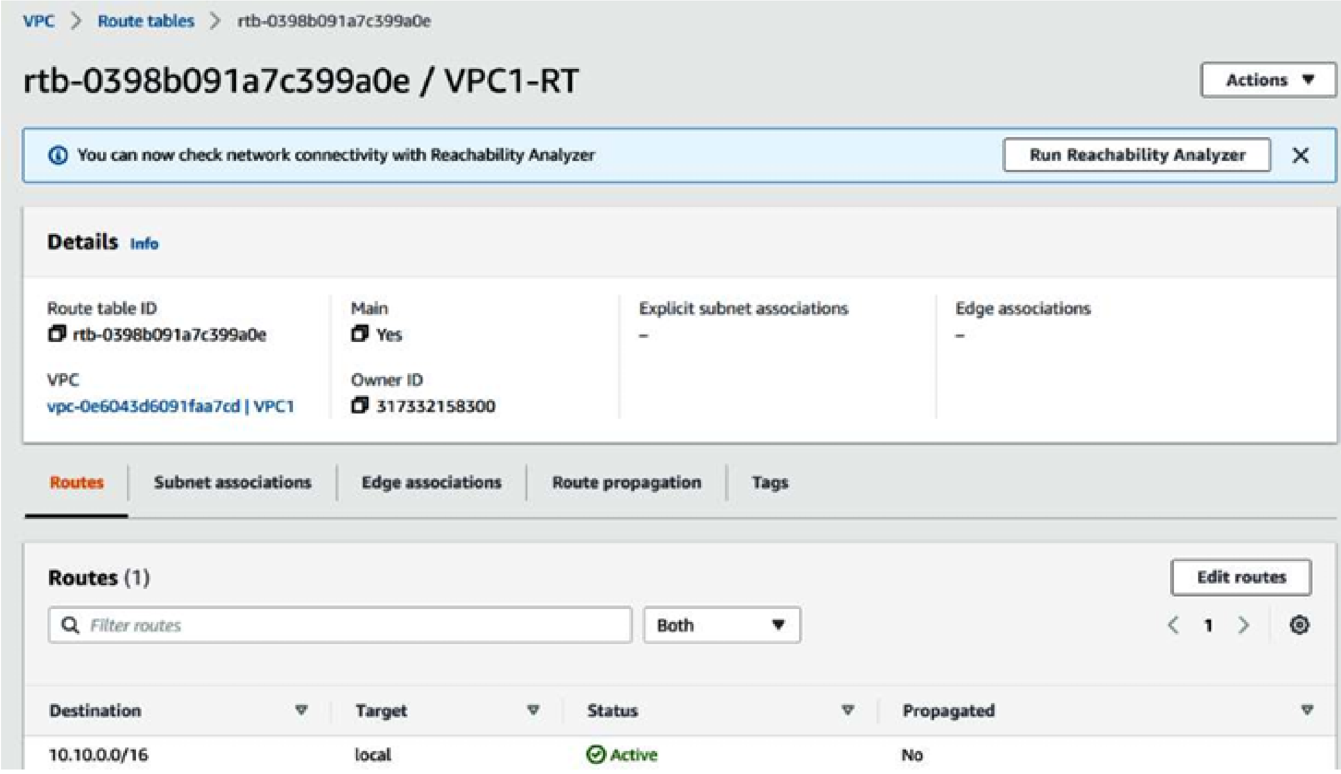


Figure 3.11 – Route table for VPC1

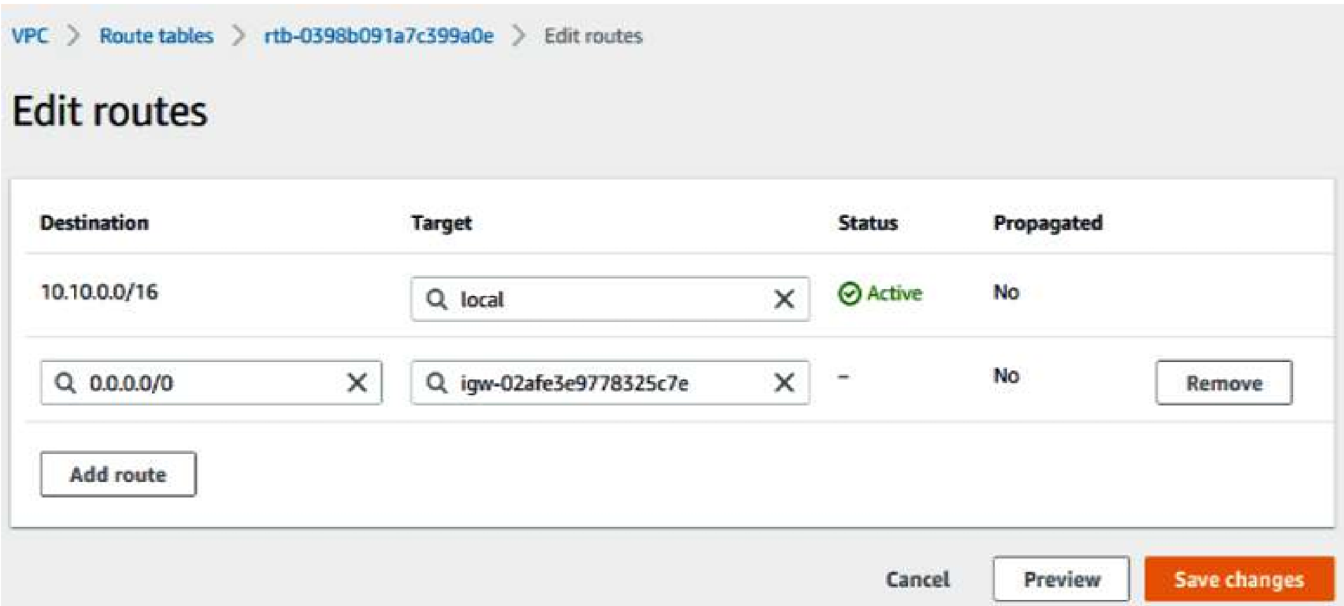


Figure 3.12 – Add internet route to route table

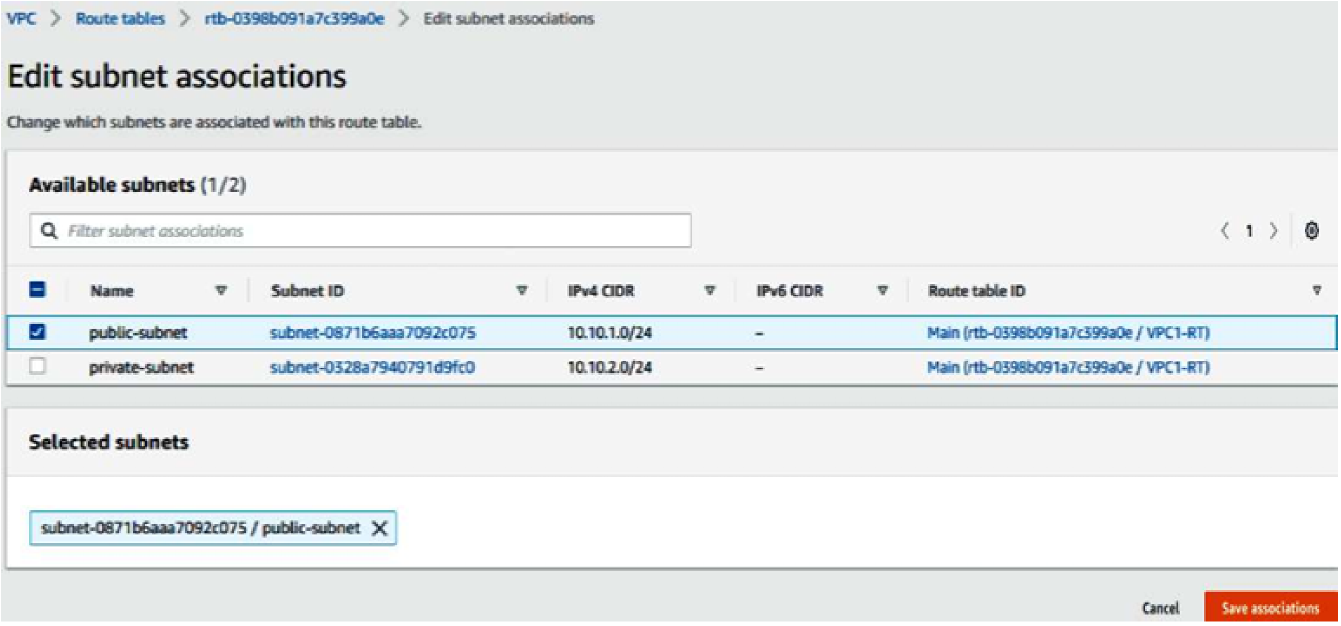


Figure 3.13 – Associate route table to subnet

3. SSH Access

SSH into the public EC2 instance.

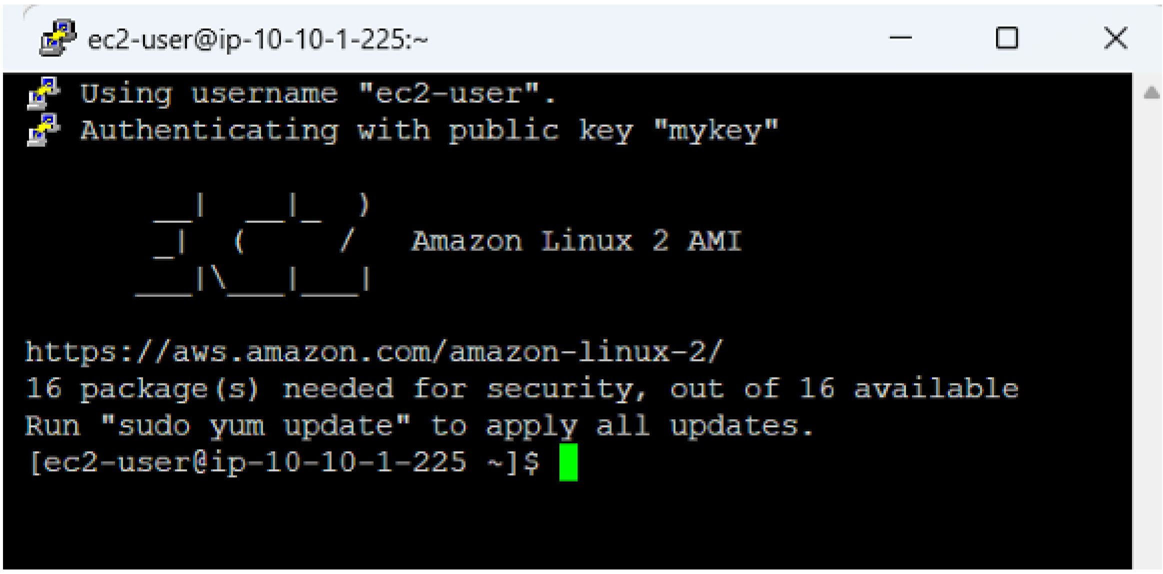


Figure 3.14 – SSH to EC2-1 from the internet

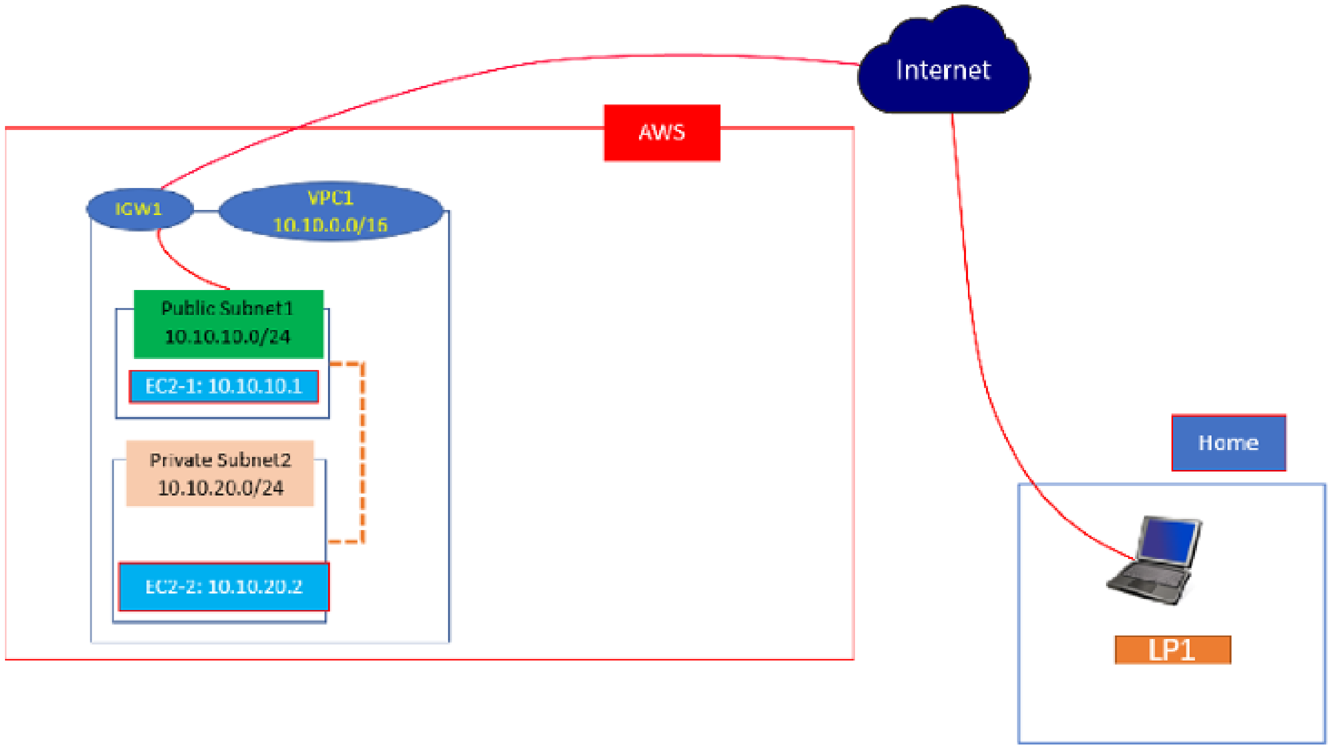


Figure 3.15 – AWS cloud with internet access

Part Two – Connecting Multiple VPCs

Additional VPCs can be connected using **VPC Peering**.

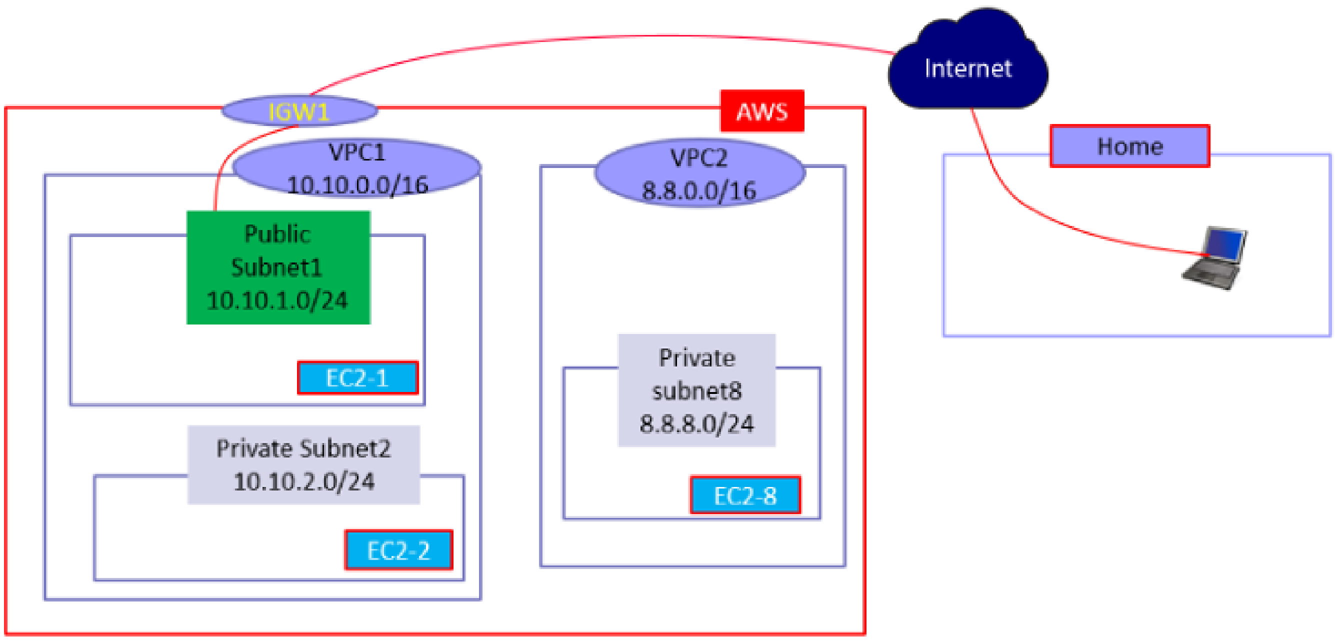


Figure 3.16 – VPC network architecture

Peering the VPCs

Create a peering connection between VPC1 and VPC2.

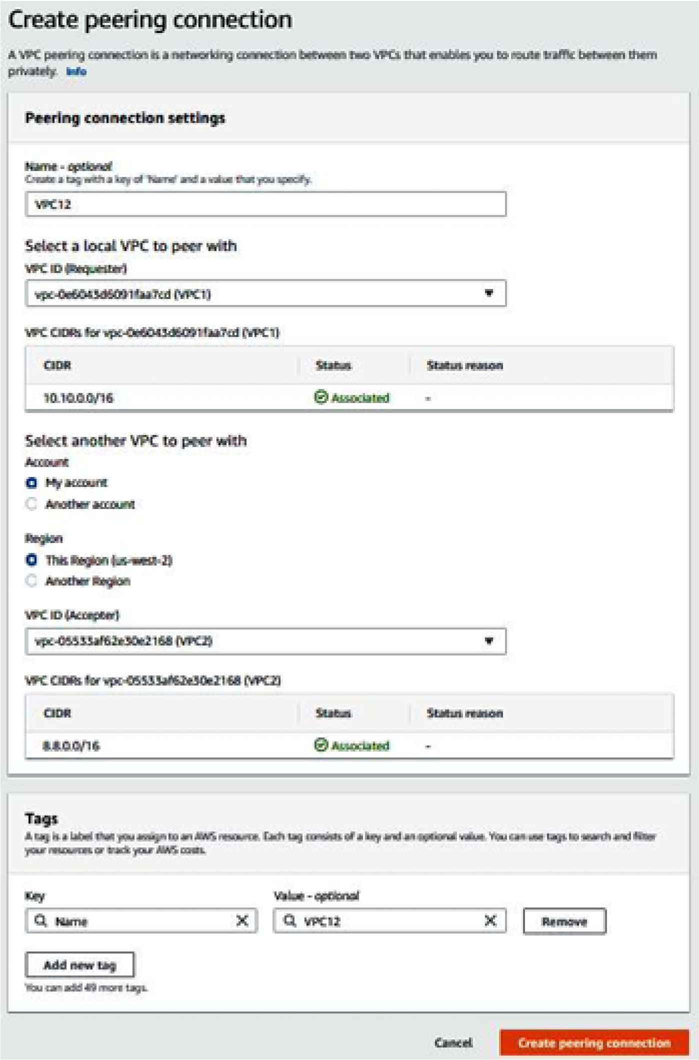


Figure 3.17 – VPC peering

Update route tables on both VPCs.

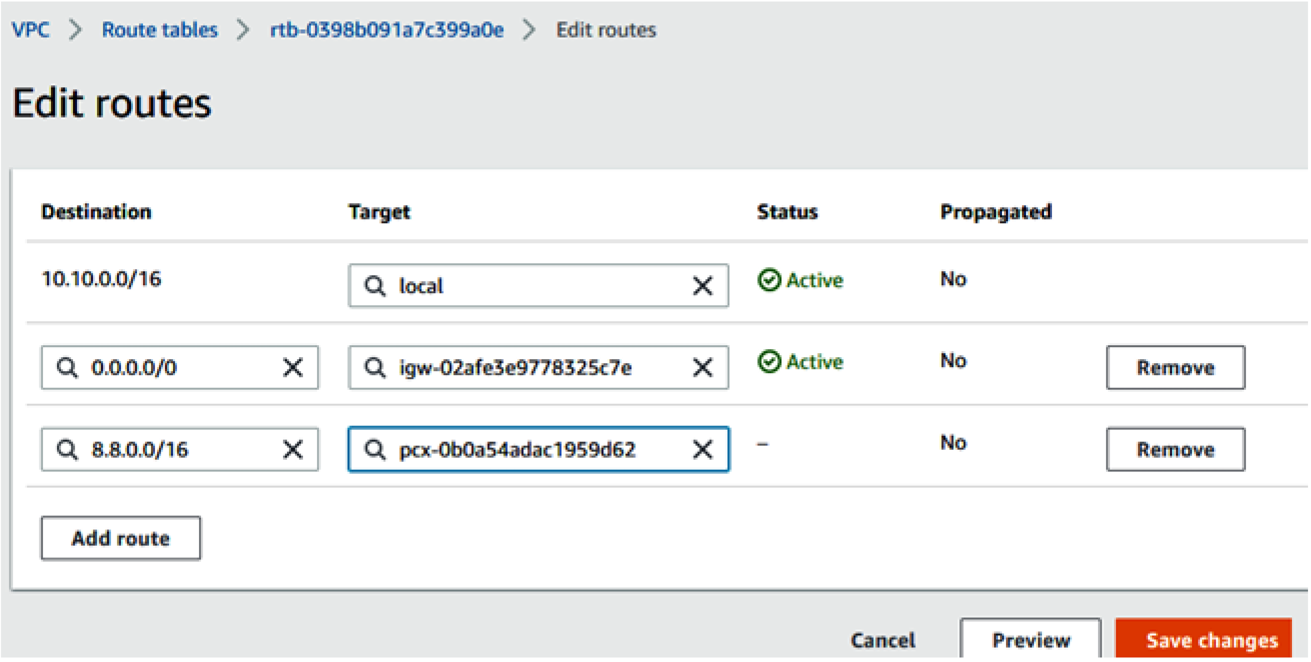


Figure 3.18 – Edit routes for the VPC1 route table

Update security groups to allow ICMP.

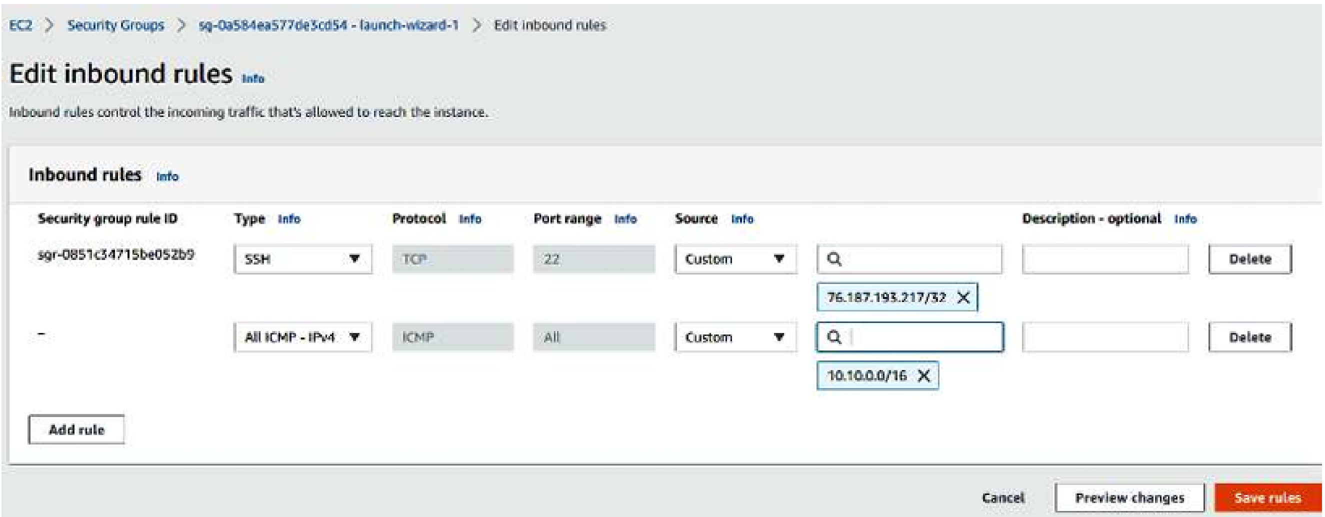


Figure 3.19 – Edit SG inbound rules for EC2-8

```
[ec2-user@ip-10-10-1-225 ~]$ ping  8.8.8.43
PING 8.8.8.43 (8.8.8.43) 56(84) bytes of data.
64 bytes from 8.8.8.43: icmp_seq=1 ttl=255 time=0.955 ms
64 bytes from 8.8.8.43: icmp_seq=2 ttl=255 time=0.553 ms
64 bytes from 8.8.8.43: icmp_seq=3 ttl=255 time=0.495 ms
^C
--- 8.8.8.43 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2023ms
rtt min/avg/max/mdev = 0.495/0.667/0.955/0.206 ms
```

Figure 3.20 – Ping EC2-8 from EC2-1

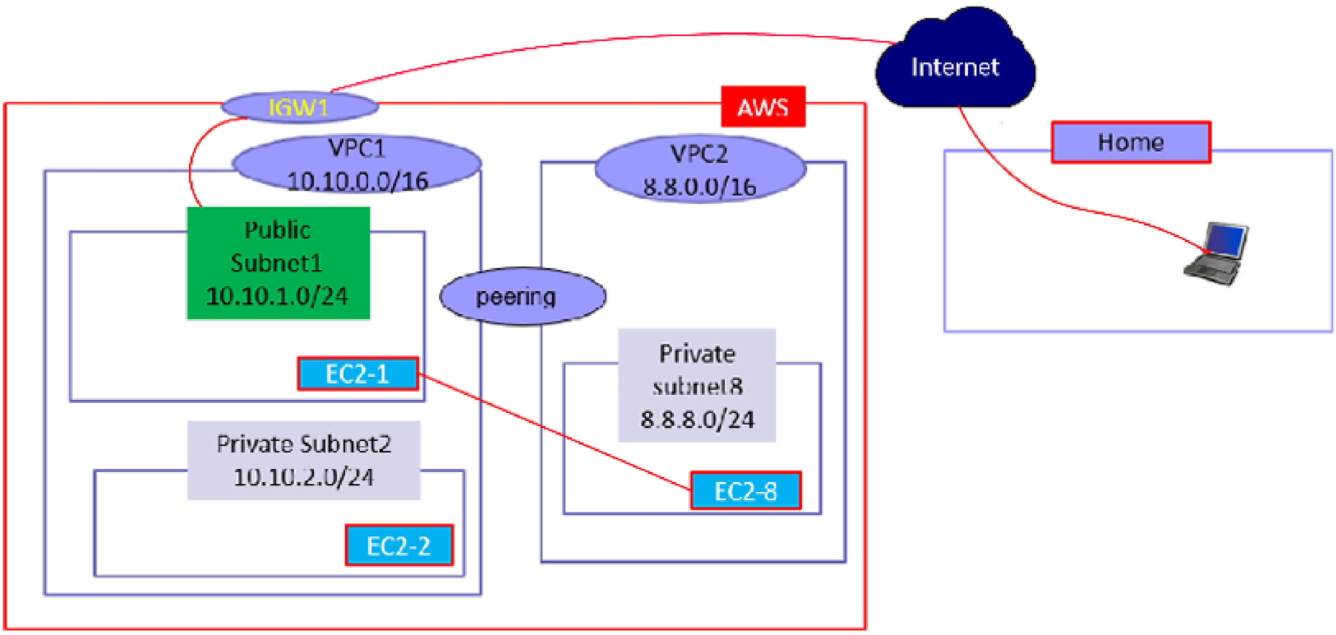


Figure 3.21 – VPC network with peering

Creating a NAT Gateway

A **NAT Gateway** enables outbound internet access for private subnet instances.

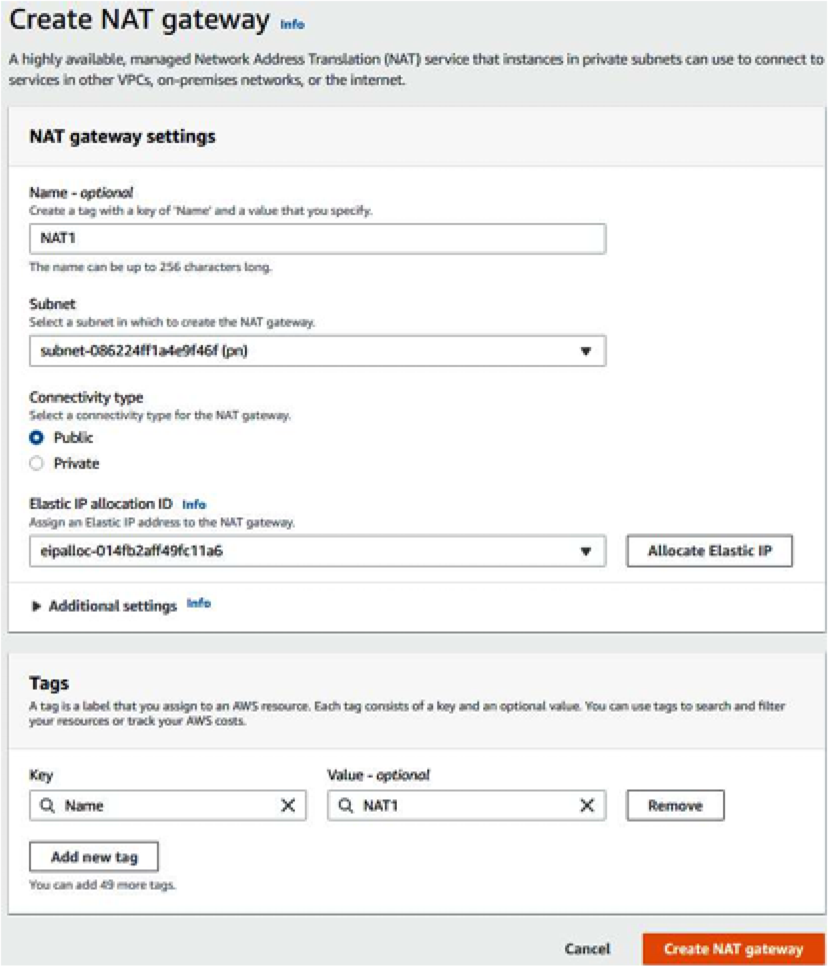


Figure 3.22 – Create NAT gateway

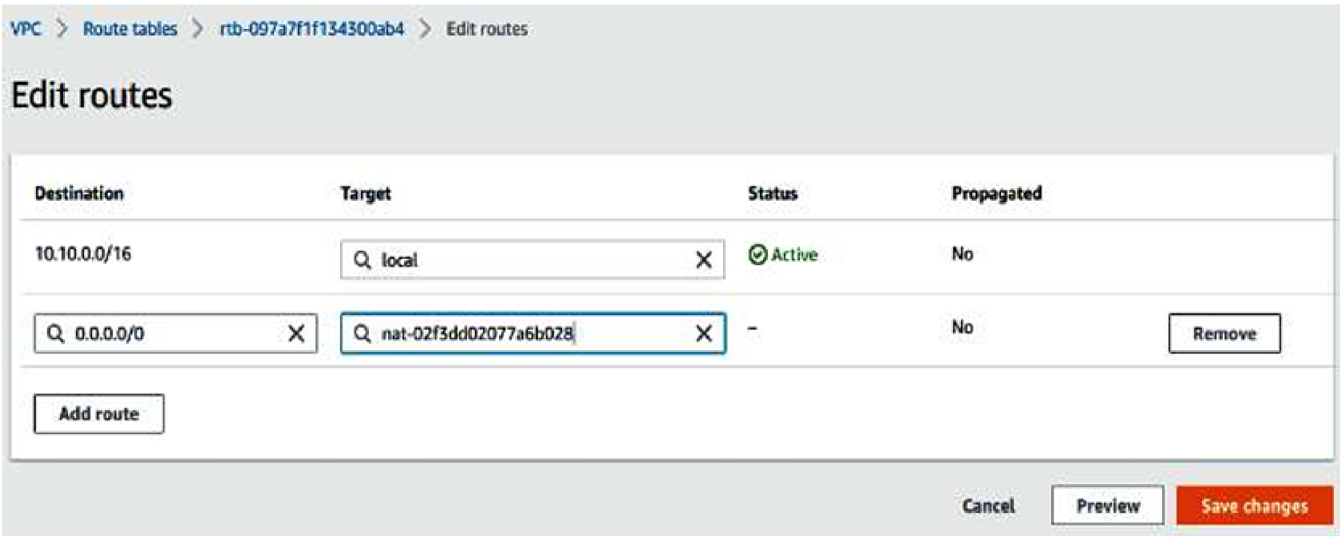
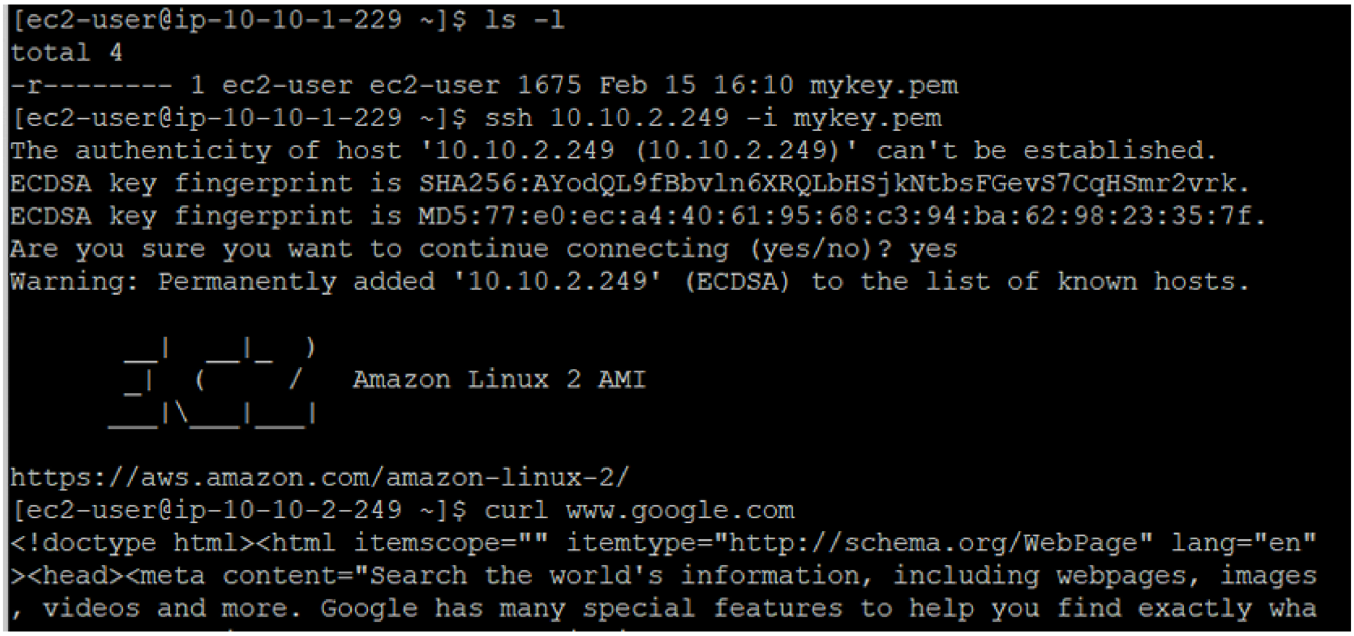


Figure 3.23 – Edit routes for the private subnet route table



```
[ec2-user@ip-10-10-1-229 ~]$ ls -l
total 4
-r-------- 1 ec2-user ec2-user 1675 Feb 15 16:10 mykey.pem
[ec2-user@ip-10-10-1-229 ~]$ ssh 10.10.2.249 -i mykey.pem
The authenticity of host '10.10.2.249 (10.10.2.249)' can't be established.
ECDSA key fingerprint is SHA256:AYodQL9fBbvln6XRQLbHSjkNtbsFGevS7CqHSmr2vrk.
ECDSA key fingerprint is MD5:77:e0:ec:a4:40:61:95:68:c3:94:ba:62:98:23:35:7f.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '10.10.2.249' (ECDSA) to the list of known hosts.

       __|  __|_  )
       _|  (     /   Amazon Linux 2 AMI
      ___|\___|___|

https://aws.amazon.com/amazon-linux-2/
[ec2-user@ip-10-10-2-249 ~]$ curl www.google.com
<!doctype html><html itemscope="" itemtype="http://schema.org/WebPage" lang="en"
><head><meta content="Search the world's information, including webpages, images
, videos and more. Google has many special features to help you find exactly wha
```

Figure 3.24 – Access www.google.com from EC2-2

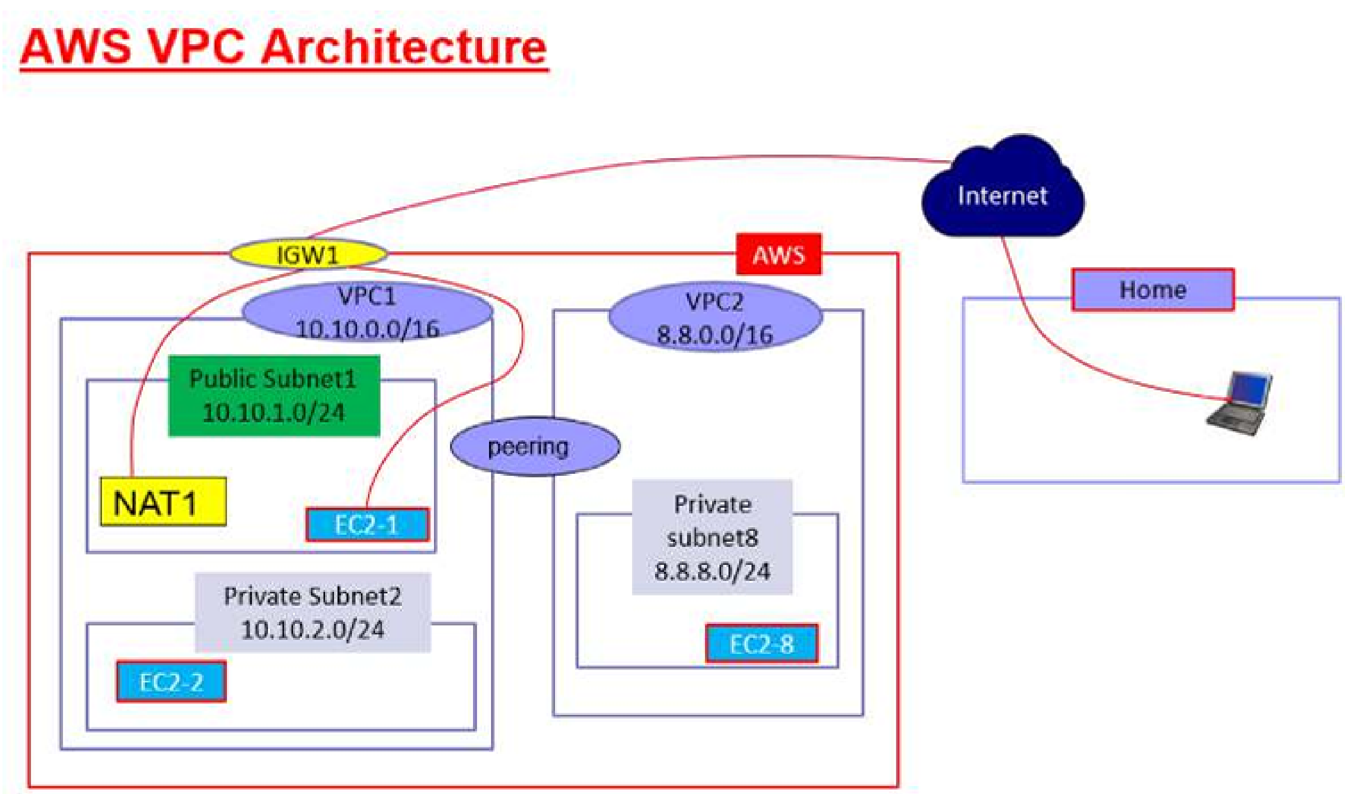


Figure 3.25 – AWS VPC architecture

Part Three – Hardening AWS Network Security

VPC Firewalls

Security Groups (SGs) are stateful, instance-level firewalls.

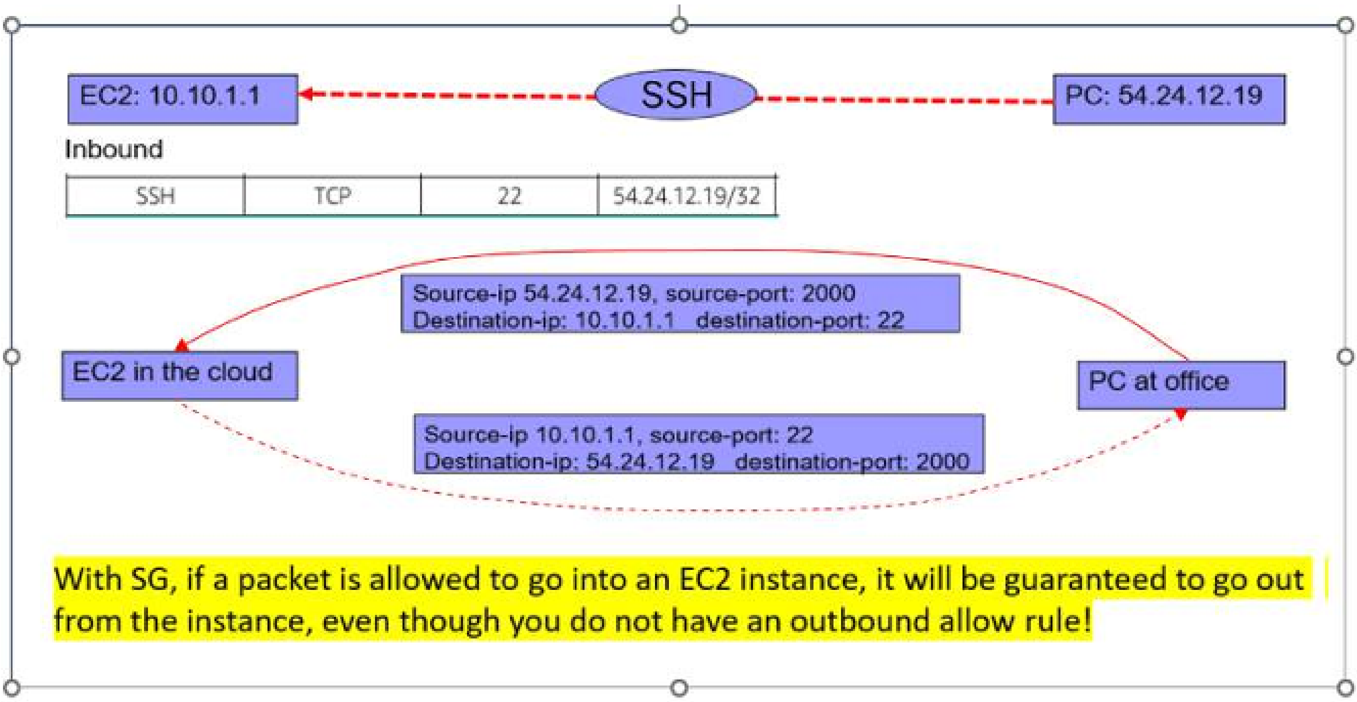


Figure 3.26 – SG is stateful (1)

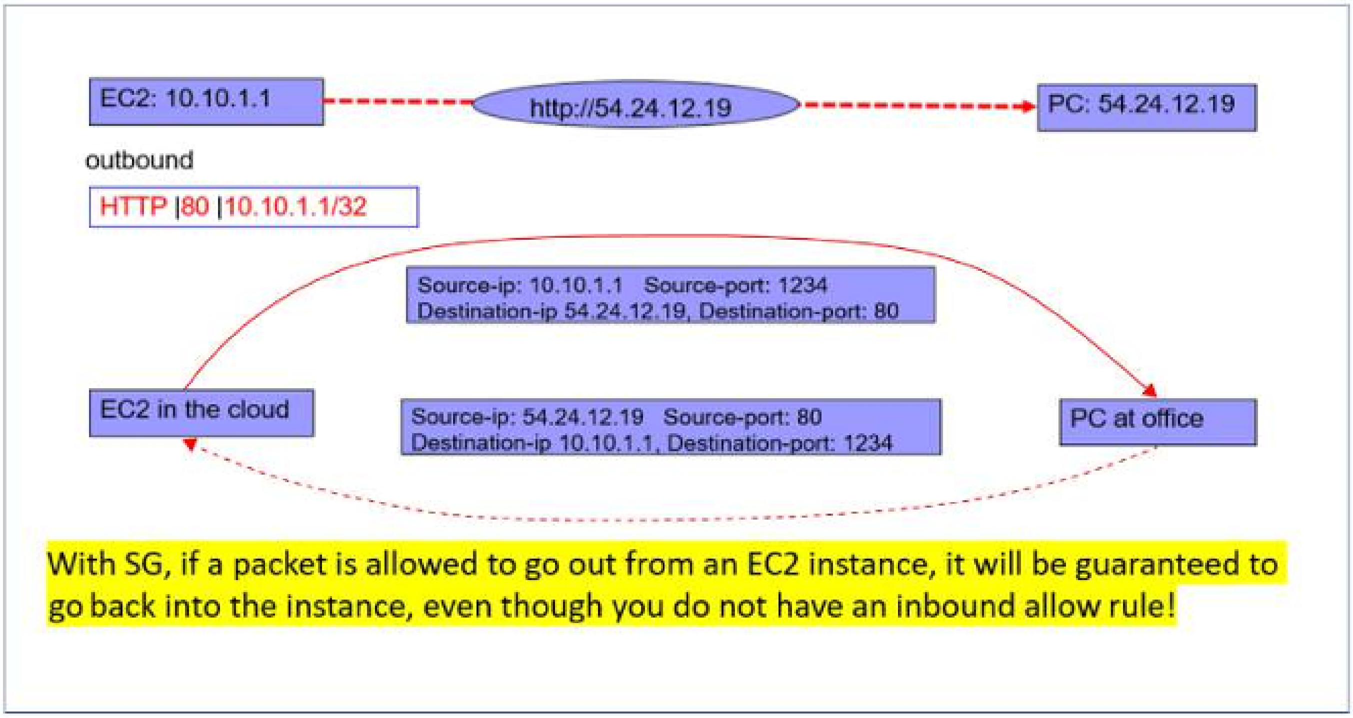


Figure 3.27 – SG is stateful (2)

Network ACLs (NACLs) are stateless, subnet-level firewalls.

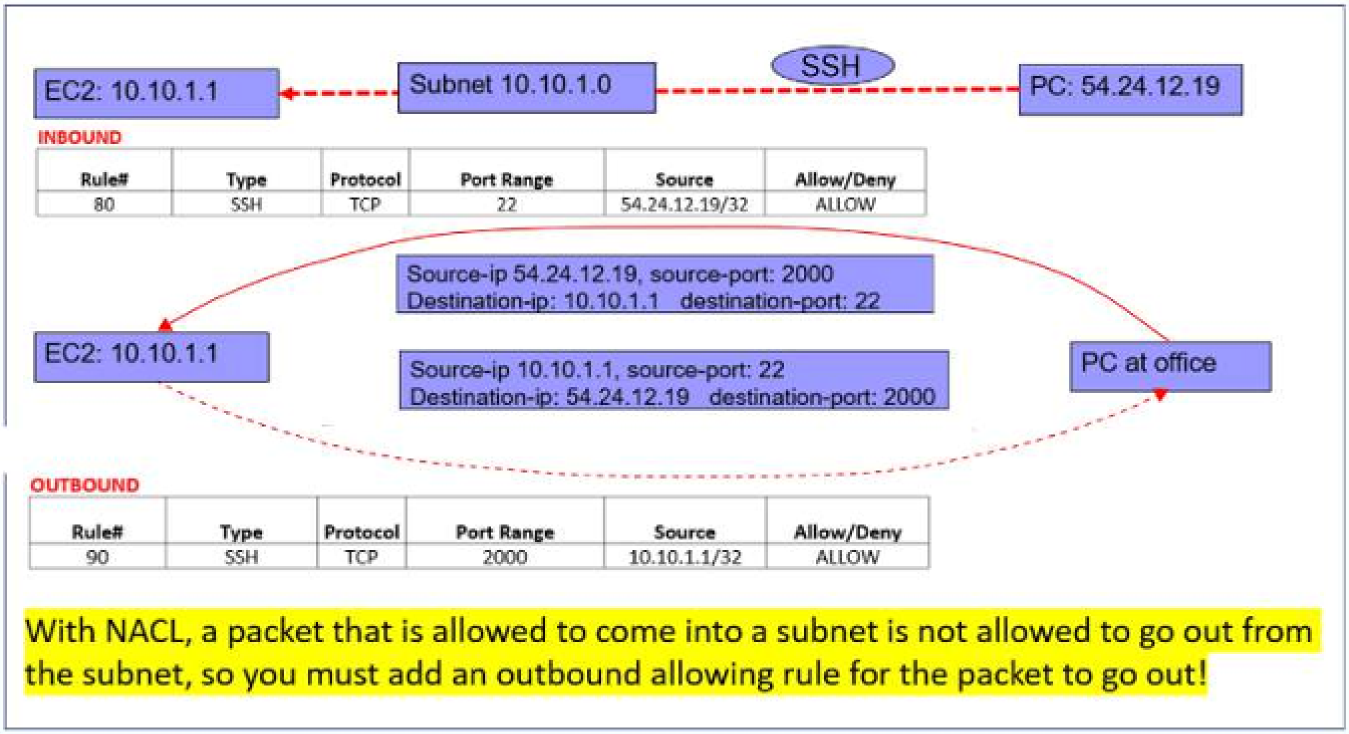


Figure 3.28 – NACL is stateless (1)

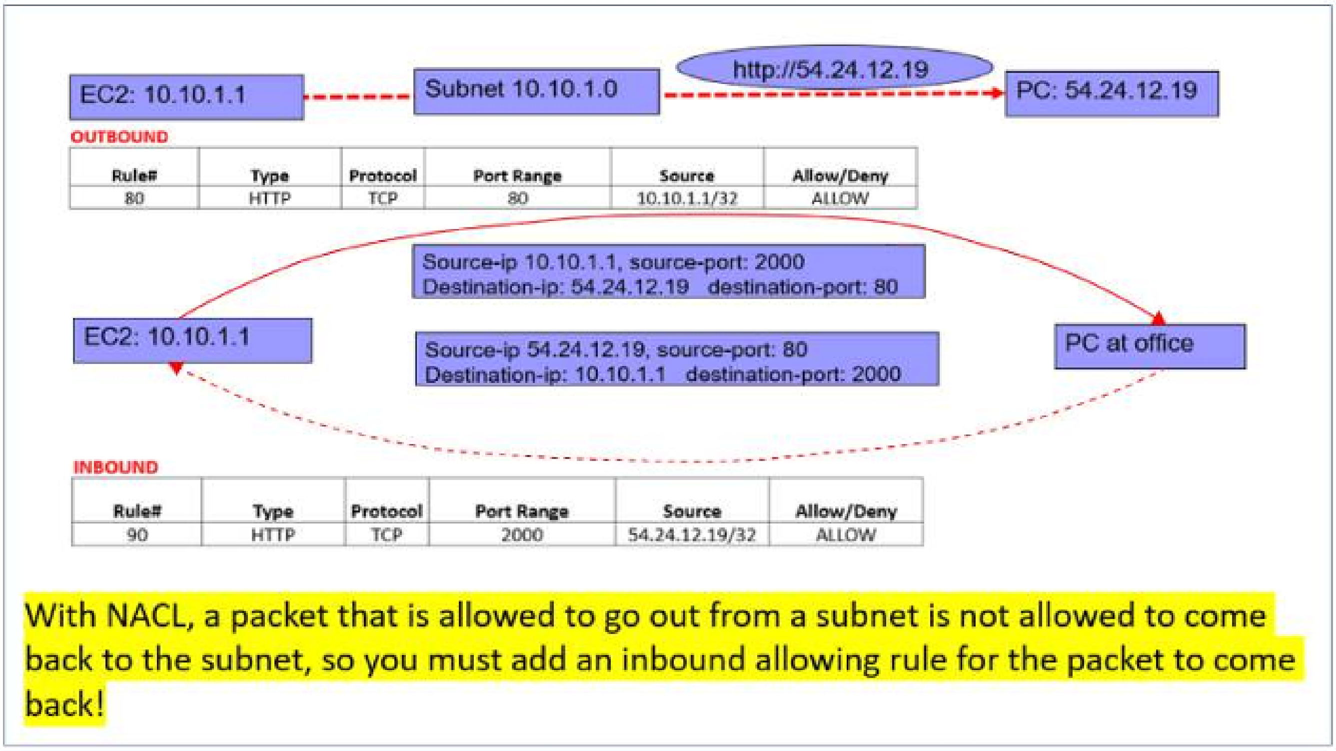


Figure 3.29 – NACL is stateless (2)

VPC Endpoints

VPC endpoints allow private access to AWS services (such as S3) without internet exposure.

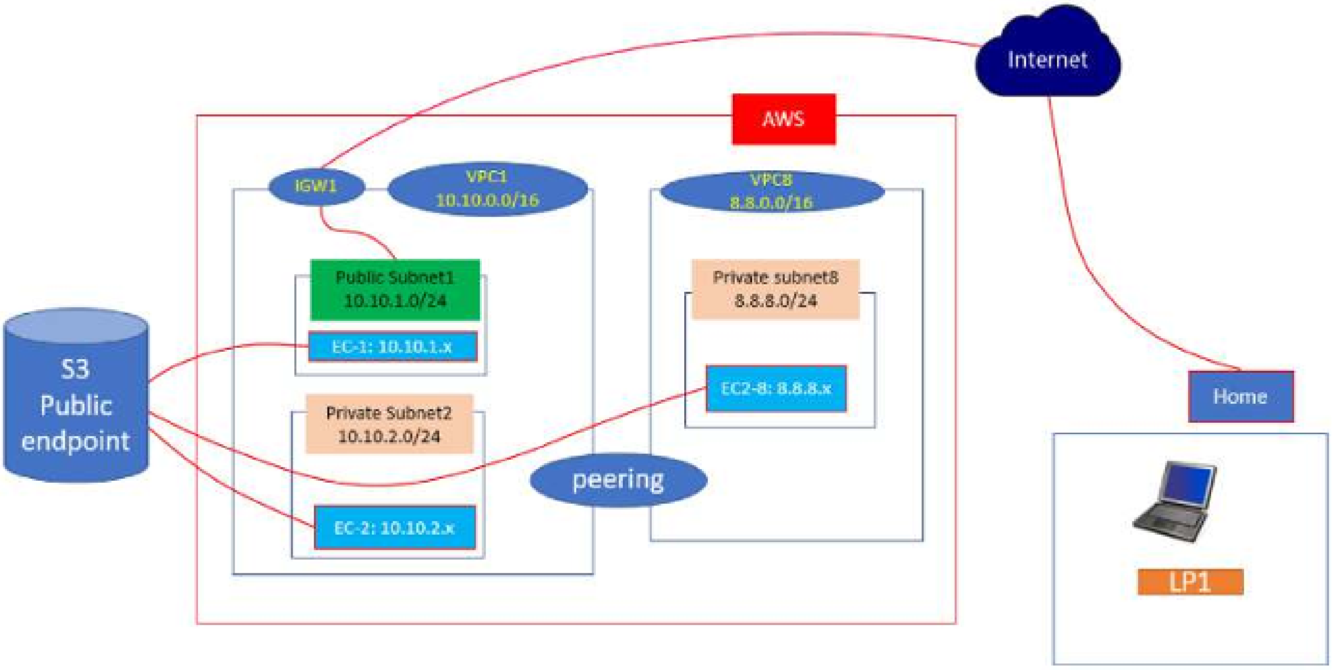


Figure 3.30 – VPC network diagram

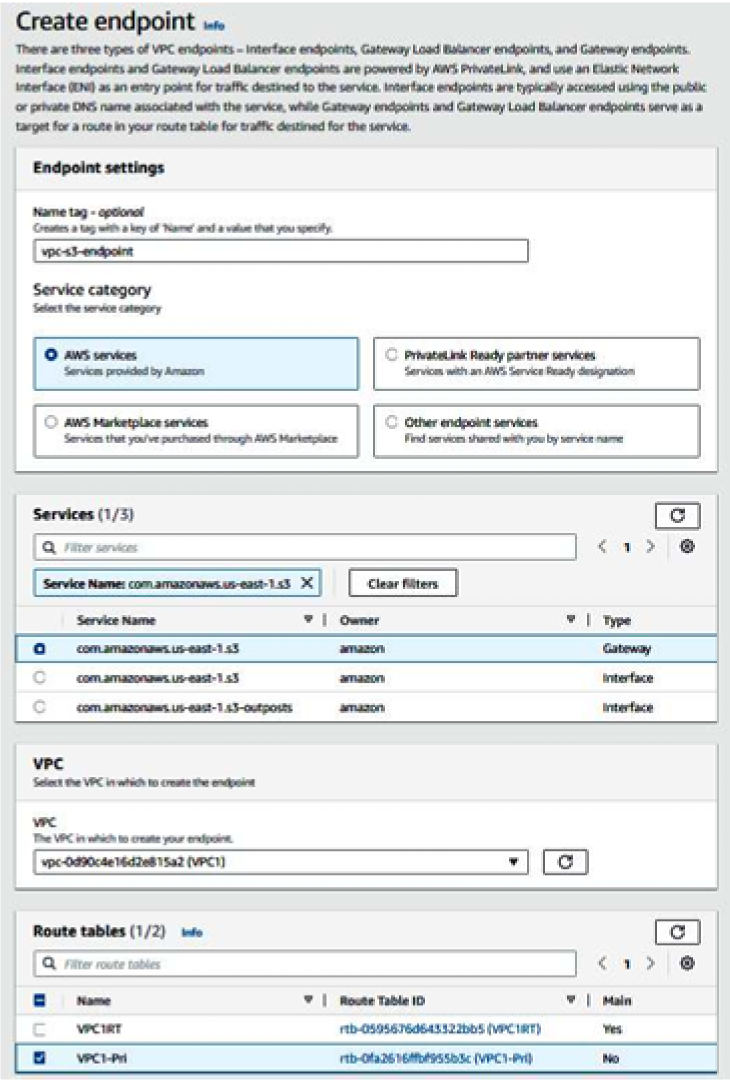


Figure 3.31 – VPC endpoint settings

Understanding Amazon Direct Connect

AWS Direct Connect provides a private, dedicated connection between on-premises data centers and AWS with predictable performance.

Setup steps include requesting a connection, selecting a partner, configuring routers, and validating connectivity.

Understanding Amazon DNS – Route 53

Amazon Route 53 is a scalable DNS service offering:

- Domain name resolution
- Traffic routing
- Health checks and failover

Supported routing policies include simple, weighted, latency-based, geolocation, failover, and multivalue routing.

Understanding the Amazon CDN

Amazon CloudFront is AWS's global CDN service, providing secure, low-latency delivery of static and dynamic content through worldwide edge locations.

Essential Commands Explained

Command	Purpose
`aws ec2 create-vpc`	Creates a new VPC with a specified CIDR block
`aws ec2 create-subnet`	Creates a subnet within a VPC
`aws ec2 run-instances`	Launches EC2 instances
`aws ec2 create-internet-gateway`	Creates an Internet Gateway
`aws ec2 create-nat-gateway`	Enables outbound internet access for private subnets
`ping`	Tests network connectivity
`ssh`	Securely connects to EC2 instances
`curl`	Tests outbound internet access

Reference Links

- AWS VPC: https://aws.amazon.com/vpc/
- AWS EC2: https://aws.amazon.com/ec2/

- AWS Direct Connect: https://docs.aws.amazon.com/directconnect/latest/UserGuide/getting_started.html
- VPN over Direct Connect: https://aws.amazon.com/premiumsupport/knowledge-center/create-vpn-direct-connect/
- Amazon Route 53: https://aws.amazon.com/route53/
- Amazon CloudFront: https://aws.amazon.com/cloudfront/
